

to hacking attacks. They showed how hackers can kill a car engine remotely, turn off the brakes to prevent the car from stopping and make instruments give false readings.

The team concentrated its attacks on the electronic control units (ECUs) scattered throughout the vehicles that oversee the workings of many car components. It is thought that new generation of vehicles have about 100 MB of binary code spread across up to 70 ECUs.

Individual control units typically oversee one sub-system but ECUs communicate so that many different systems can be controlled as the situation demands. The team created a software called CarShark to monitor communications between the ECUs and insert fake packets of data to carry out attacks. It got access to the ECUs via communication ports fitted as standard on most cars that enable mechanics to gather data about a vehicle before they begin servicing or repair work.

Self-assembling optical computer

North Carolina-based Duke University researchers have used DNA to create self-assembling logic circuits. These molecules make use of light for binary computation and communication.

The researchers mixed customised snippets of DNA and other molecules to create billions of identical, nanometre-scale waffle-shaped structures. These structures are well-known for their ability to absorb and emit light of different wavelengths. It is this ability to process input data that the researchers sought to exploit as a digital process.

The researchers used a conventional computer to determine which DNA fragments will preferentially clip together to form the functioning logic. Instead of conventional circuits using a flow of electrons to switch 1's and 0's, light can be used to effect similar responses from DNA-based switches much faster, claim the researchers.

Smallest transistor paves way for superfast computers

An international team of scientists claims to have created the world's smallest transistor from a quantum dot of just seven atoms. The tiny transistor, which is a mere four billionths of a metre long, may pave the way for superfast computers, scientists believe.

According to the scientists, this transistor can be used to regulate and control electrical current flow like a commercial transistor but it represents a key step into a new age of atomic-scale miniaturisation and superfast computers.

"The significance of this achievement is that we are not just moving atoms around or looking at them through a microscope. We are manipulating individual atoms and placing them with atomic precision, in order to make a working electronic device," said a team member.

Multitouch makes typing from touchscreens easier

To make typing long documents from touchscreens easier and faster, a French



Great Wall Gbook tablet that uses Stantum's digital-resistive multitouch technology

company called Stantum has developed the digital-resistive multitouch technology. The technology includes controllers and sensors so that input into the touchscreen – so-called soft keyboard – is reflected more quickly and accurately on the screens.

The technology is being licenced to touch-panel makers like STMicroelectronics. One company, Great Wall, has already displayed a tablet called Gbook that utilises Stantum's technology.

The technology measures the location and force of screen contact to better filter out unintentional touches. It can recognise input from up to ten fingers simultaneously. That removes hindrances you would otherwise face in writing long-form documents on a soft keyboard and helps gamers who want fast response rates and high performance from computers.

Wi-Di for handhelds

Intel is all set to integrate another path-breaking technology called Wi-Di into handheld devices like mobiles and netbooks. The Wi-Di technology is the wireless display technology that



sparsh
dependable security solutions
for developing tomorrow

Made in India

540 TVL Camera

- High Resolution 540 TVL Camera
- Sony Super HAD-II CCD
- Better Night Vision